SmartRoute Pro — Comprehensive Academic Project Report

Integrated Mini-Project across Analysis and Design of Algorithms (ADA), Statistical Data Analysis (SDA) & Cloud Computing (CC)

1. Abstract

SmartRoute Pro is a unified logistics route optimization and data analytics platform designed to solve urban consignment delivery challenges. The system combines graph-theoretic shortest path routing (Dijkstra's Algorithm with a binary min-priority queue, achieving O((V + E) log V) time complexity), spatial clustering (Equirectangular local metric projected K-Means with Silhouette evaluation), inferential statistics (Bessel-corrected sample variance, Pearson correlation, and Ordinary Least Squares linear regression), and cloud-native DevOps principles (multi-stage Docker containerization, Gunicorn WSGI, managed PostgreSQL connection pooling, and automated GitHub Actions CI/CD pipelines). This report documents the mathematical methodologies, architectural decisions, empirical benchmark results, and verification testing for the complete platform.

2. Problem Statement & Objectives

Modern urban logistics systems suffer from three intertwined bottlenecks: (1) inefficient vehicle dispatching along sub-optimal road corridors, (2) unpartitioned customer locations causing overlapping delivery zones, and (3) lack of operational health observability across the supporting server infrastructure. The objective of this project is to create an integrated, full-stack platform addressing each dimension with mathematical rigour, statistical validation, and production-grade cloud deployment standards.

3. Technology Stack & Multi-Tier Architecture

Tier	Technology Stack	Curricular Alignment
Presentation Tier	React 19, TypeScript 5.8, Vite, Tailwind CSS, Recharts, Lucide Icons	Frontend Engineering & UI/UX Design
Application Tier	Python 3.11, Flask 3.0, Gunicorn 26.2 (gthread WSGI), ReportLab 5.0	Core Application Gateway & Services
Algorithm Services (ADA)	Adjacency List Graph, Python heapq Min-Priority Queue, Step Tracer	**Analysis & Design of Algorithms (ADA)**
Analytics Services (SDA)	Scikit-Learn KMeans, Scipy Stats OLS Regression, Metric Projection	**Statistical Data Analysis (SDA)**
Cloud & Persistence (CC)	Docker (Multi-Stage), Managed PostgreSQL 16, Alembic Migrations, GitHub Actions	**Cloud Computing (CC)**

4. ADA Methodology: Dijkstra Shortest Path Engine

Given a directed network G = (V, E) loaded from relational Location and Road tables, Dijkstra's algorithm computes the single-source shortest path using a min-heap where each heap element is (distance, node_id). To support directed edge constraints, edges are loaded conditionally based on is_active = True. Predecessor mapping reconstructs the optimal vertex sequence upon reaching the destination.

Empirical ADA Telemetry: Total Stored Route Calculations: 2 • Mean Execution Time: 0.07 ms • Latest Target Distance: 0.0 km.

5. SDA Methodology: Spatial Clustering & Inferential Statistics

Locations are clustered in ground kilometers via equirectangular metric projection: x = (lon - mean_lon) * 111.320 * cos(mean_lat), y = (lat - mean_lat) * 110.574. Clustering quality is evaluated using the Silhouette Coefficient s = (b - a) / max(a, b) and the Elbow Method across K = 1..N. Inferential statistics verify that road transit duration scales linearly with road distance (evaluated via OLS linear regression).

6. CC Methodology: Cloud Architecture, Containerization & CI/CD

The platform implements production-ready Docker containers with non-root security (UID 1001), multi-stage Nginx builds (<25MB), fail-closed production PostgreSQL enforcement (no SQLite fallback permitted in production), and end-to-end GitHub Actions continuous integration automating 55+ test cases.

7. Quality Assurance & Automated Test Results

Test Module	Scope & Test Cases Covered	Execution Result
test_health.py	API root, health endpoint, version & module verification (3 tests)	**3 / 3 PASSED**
test_phase2.py	Relational models, Delivery CRUD, foreign keys, cascade deletes (10 tests)	**10 / 10 PASSED**
test_dijkstra.py	Graph theory, priority queue, directed roads, edge cases (15 tests)	**15 / 15 PASSED**
test_sda.py	K-Means, projection, silhouette, Bessel variance, OLS regression (16 tests)	**16 / 16 PASSED**
test_cloud.py	Cloud probes, latency, no SQLite fallback, security headers, auth (11 tests)	**11 / 11 PASSED**
Total Backend Suite	**Unified Automated Test Suite across all curricular domains**	**55 / 55 PASSED (100%)**

8. Known Limitations & Future Scope

Limitations: Dijkstra solves single-pair shortest paths but does not address multi-stop combinatorial TSP with vehicle capacity constraints. K-Means assumes spherical cluster geometries and requires K to be designated a priori. Cloud deployment is container-ready but awaits user-approved external account provisioning.
Future Scope: Incorporate genetic algorithms or simulated annealing for full multi-vehicle TSP with time windows (VRPTW), support PostGIS spatial indexing directly in SQL, and configure Redis caching for pre-computed graph shortest paths.